

MSE 360: Kinetic Processes in Materials

Homework #4

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. A fuel cell is provided with a molar flux of H_2 , $J_{H_2} = 10^{-5} \text{ mol } H_2/(\text{cm}^2 \cdot \text{s})$. Convert this molar H_2 flux into the following sets of units, given $T = 300 \text{ K}$, $P = 1 \text{ atm}$. Assume H_2 behaves as an ideal gas.
- Volumetric flux [$\text{cm}^3/(\text{cm}^2 \cdot \text{s})$]
 - Mass flux [$\text{g}/(\text{cm}^2 \cdot \text{s})$]
 - Molecular flux [molecules/ $(\text{cm}^2 \cdot \text{s})$]
 - Current density [A/cm^2], by assuming each hydrogen molecule is oxidized in a fuel cell to produce 2 electrons

a) volumetric flux [$\text{cm}^3/(\text{cm}^2 \cdot \text{s})$]

$$PV = nRT$$

$$V_{m,H_2} = \frac{V}{n} = \frac{RT}{P} = \frac{8.314 \text{ J/mol} \cdot \text{K} \cdot 300 \text{ K}}{1 \text{ atm} \cdot 101,300 \text{ Pa/atm}}$$

$$\begin{aligned} V_{m,H_2} &= 2.462 \times 10^{-2} \text{ m}^3/\text{mol} \\ &= 24,620 \text{ cm}^3/\text{mol} \end{aligned}$$

$$\begin{aligned} J_{\text{vol},H_2} &= J_{\text{mol},H_2} V_{m,H_2} \\ &= 10^{-5} \frac{\text{mol } H_2}{\text{cm}^2 \cdot \text{s}} \times \frac{24,620 \text{ cm}^3}{\text{mol}} \end{aligned}$$

$$J_{\text{vol},H_2} = 0.246 \text{ cm}^3/\text{cm}^2 \cdot \text{s}$$

b) mass flux [$\text{g}/(\text{cm}^2 \cdot \text{s})$]

$$J_{\text{mass}, \text{H}_2} = J_{\text{mol}, \text{H}_2} \cdot M_{\text{H}_2}$$
$$= \frac{10^{-5} \text{ mol}}{\text{cm}^2 \cdot \text{s}} \times \frac{2.016 \text{ g}}{\text{mol}}$$

$$J_{\text{mass}, \text{H}_2} = 2.02 \times 10^{-5} \text{ g}/\text{cm}^2 \cdot \text{s}$$

c) molecular flux [$\text{molecules}/\text{cm}^2 \cdot \text{s}$]

$$J_{N, \text{H}_2} = J_{\text{mol}, \text{H}_2} N_A$$

$$J_{N, \text{H}_2} = \frac{10^{-5} \text{ mol}}{\text{cm}^2 \cdot \text{s}} \times \frac{6.022 \times 10^{23}}{1 \text{ mol}}$$

$$J_{N, \text{H}_2} = 6.02 \times 10^{18} \text{ molecules}/\text{cm}^2 \cdot \text{s}$$

d) current density [A/cm^2]

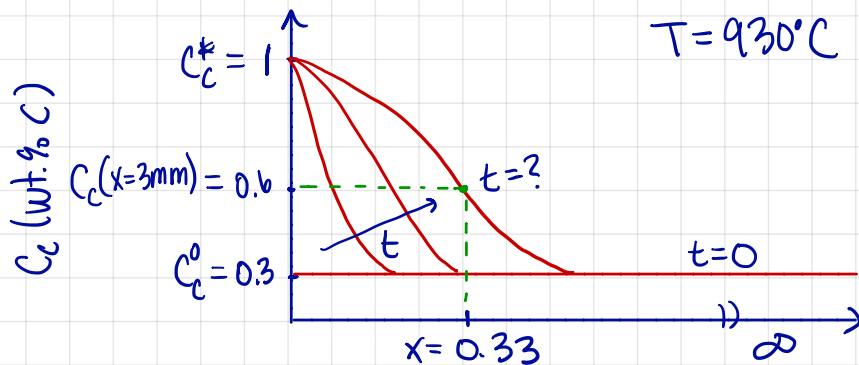
$$i = \frac{10^{-5} \text{ mol}}{\text{cm}^2 \cdot \text{s}} \times \frac{6.02 \times 10^{23}}{\text{mol H}_2} \times \frac{2e}{\text{H}_2} \times \frac{1.602 \times 10^{-19} \text{ C}}{e^-}$$

$$i = 1.93 \text{ A}/\text{cm}^2$$

2. The aim of carburization is to increase the carbon concentration in the surface layers of steel in order to achieve a harder wear-resistant surface. This is usually done by holding the steel in a gas mixture containing CH_4 and/or CO at a temperature where austenite is present. A thick plate of 0.3 wt % C steel is carburized at 930°C . The carburizing gas used in the treatment holds the surface concentration at 1.0 wt % C. Assume the plate is so thick that the diffusing carbon does not reach the opposite side of the plate over the course of the treatment.

Given that the diffusivity of carbon in iron at this temperature is $D = 1.1 \times 10^{-9} \text{ m}^2/\text{s}$, determine how long it will take for the carbon concentration to reach 0.6 wt % C at a depth of 0.33 mm.

transient semi-infinite, planar diffusion



$$C_c(x, t) = C_c^0 + (C_c^* - C_c^0) \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

$$\frac{C_c(x, t) - C_c^0}{C_c^* - C_c^0} = \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right] \quad D_{\text{C in Fe, } 930^\circ\text{C}}$$

$$\frac{0.6 - 0.3}{1 - 0.3} = 0.4286 = \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

$$\text{If } \operatorname{erfc}[w] = 0.4286, \operatorname{erf}[w] = 0.5714 \\ w \approx 0.56$$

$$\omega = \frac{x}{2\sqrt{Dt}}$$

$$t = \frac{x^2}{4\omega^2 D}$$

$$= \frac{(3.3 \times 10^{-4} \text{ m})^2}{4(0.56)^2 1.1 \times 10^{-9} \text{ m}^2/\text{s}}$$

$$t = 79 \text{ s}$$